SHANNMUKA BUDDABATHINI

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with 6+ years of experience building reliable, cloud-native platforms and automation on Azure and Kubernetes. CKA-certified and Microsoft Certified DevOps Engineer Expert with deep expertise in Python, Kubernetes, CI/CD, infrastructure as code, and observability, and a track record of leading disaster recovery for 100+ applications, supporting ~500 application teams in production, and governing 70,000+ cloud subscriptions at 99% policy compliance. Builds LLM-powered operations tooling (Azure OpenAI, RAG) that classifies alerts at ~97% accuracy, and uses GitHub Copilot, Claude, Cursor, and Walmart's Wibey and Code Puppy to accelerate development and automate complex workflows.

TECHNICAL SKILLS

Languages: Python, Java, C#, SQL

AI / LLM Tools: Copilot, Claude, Gemini, Cursor, Wibey, Code Puppy, Azure OpenAI, OpenAI API, LangChain, RAG

Cloud & DevOps: Azure (AKS, Azure Policy, PIM, Azure AD / Entra ID, Azure Monitor), AWS (EC2, RDS, S3), Docker, Kubernetes, Helm, Terraform, ARM, Bicep, Ansible

CI/CD & Build: Jenkins, Azure DevOps, Maven, JFrog, SonarQube

Platform & Operations: WCNP, OneOps, Service Registry, Service Mesh, Incident management, On-call & production support, Disaster recovery (BCDR)

Observability: Prometheus, Grafana, Dynatrace, Azure Monitor, OpenObserve, xMatters

Backend: REST APIs, Flask, Connexion, OpenAPI, API Gateway & Proxy, Rate Limiting, Microservices, Spring Boot, Kafka

Databases & OS: MySQL, Azure SQL Database, Oracle; Linux (Ubuntu, RHEL), Windows; PowerShell

PROFESSIONAL EXPERIENCE

Walmart | Bentonville, AR *May 2024 – Present*

Software Engineer III

- Cut regional-outage recovery from ~20 min to 2–3 min with BCDR YAML playbooks and one-click DR automation; led recovery of 100+ applications during an Azure South Central outage.
- Designed and built spec-first REST Alert Controller APIs (Python, Flask, Connexion, OpenAPI 3.0.3) integrated with xMatters, suppressing ~40% of alerts (~100 of ~250 per 8-hour shift) across ~500 Marketplace applications during maintenance windows.
- Integrated an LLM enrichment layer (OpenAI API, Azure OpenAI, RAG, Claude and Gemini models) into the Alert Controller to classify alerts and infer application context at ~97% accuracy, keeping false pages from reaching on-call engineers.
- Improved reliability of the Alert Controller and DR automation APIs with an API proxy/gateway, rate limiting, and throttling (MySQL, Azure SQL Database); now being implemented across Marketplace applications.
- On call for ~500 Marketplace application teams, speeding resolution of CI/CD, Kubernetes, service mesh/registry, and config issues; monitor priority T0 deployments and guide teams as needed.
- Lead incident bridges for golden-signal issues (latency, availability, 4xx, 5xx), driving root cause analysis and prevention.
- Re-platformed manual OneOps VM runbooks to WCNP using Python, CI/CD, and Kubernetes, cutting pre-owned inventory feed categorization from 1–2 weeks to ~5–10 minutes.
- Use GitHub Copilot, Claude, Cursor, Wibey, and Code Puppy daily to accelerate development and debugging.

MAK Technologies LLC | Springboro, OH *Jul 2023 – Apr 2024*

Software Developer

- Developed backend services and RESTful APIs (Java) for internal tools and Azure-hosted services, with validation and structured error handling.
- Containerized services with Docker and deployed to Azure Kubernetes Service using Helm, with readiness/liveness probes and environment-specific settings.
- Integrated Apache Kafka for asynchronous messaging between microservices, decoupling components and absorbing traffic spikes.

- Built Jenkins CI/CD pipelines running Maven builds and tests, SonarQube quality checks, Docker image builds, and Helm deployments, reducing manual release steps.
- Defined Azure infrastructure as code with Terraform and Bicep (resource groups, AKS, storage, networking) so environments were reproducible and versioned in Git.

Microsoft | Redmond, WA *Oct 2022 – Jun 2023*

Cloud Engineer

- Spearheaded an automated cloud governance solution for the Security Intelligence & Engineering (SI&E) division that onboarded a tenant with 70,000+ subscriptions, reducing deployment costs by 15%.
- Built backend Java REST APIs to orchestrate Azure custom and built-in policies and send notification events to Microsoft Teams and Outlook.
- Authored custom Azure Policies alongside built-in policies to reach 99% compliance across all Azure resources in the Microsoft tenant, resulting in a 50% reduction in security incidents.
- Implemented Privileged Identity Management (PIM) in Azure custom policies to prevent permanent privileged access, plus an Azure AD (B2B/B2C) IAM framework with Conditional Access (SAW, SAS-ALT).
- Architected an Azure DevOps pipeline, authenticated with Service Principals, to deploy Azure policies and Java and C# .NET applications across multiple tenants, cutting policy rollout time by 50%.
- Automated 80% of deployment tasks with ARM templates and Terraform, and wrote PowerShell scripts to create and delete Azure resources and publish policy files to file shares.

MAK Technologies LLC | Springboro, OH *Aug 2022 – Sep 2022*

DevOps Engineer Intern

- Gathered client requirements and rolled out Azure Privileged Identity Management (PIM), decreasing unauthorized access by 30% and improving security compliance by 25%.
- Provisioned and configured client virtual machines with Ansible playbooks and Terraform modules.

ZENQ | Hyderabad, India *Aug 2019 – Aug 2021*

Azure Cloud Engineer

- Developed Java / Spring Boot RESTful APIs for the eVestment (Nasdaq) analytics platform for investment workflows.
- Applied Azure Policies across governance, security, and compliance work, achieving 99% compliance across all resources and reducing audit findings by 30%.
- Protected 100+ critical workloads with Azure Backup and Azure Monitor, achieving a 99% recovery rate from outages and proactively addressing 90% of potential issues before escalation.
- Authored and managed 30+ ARM templates and Terraform scripts, standardizing configurations across environments and reducing deployment times by 25%.

VVIT | Guntur, India *Jul 2018 – May 2019*

Research Engineer

- Applied ensemble machine learning (Random Forest, J48, C4.5, Naïve Bayes) and SQL to Electronic Health Record (EHR) data to support early liver cancer detection research, resulting in a co-authored international publication.

EDUCATION

M.S. Business Analytics & Information Systems — University of South Florida | Tampa, FL *Dec 2022*

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University | Guntur, India *May 2019*

CERTIFICATIONS

Certified Kubernetes Administrator (CKA), The Linux Foundation | Credential ID: LF-pywqcnzqbj

Microsoft Certified: DevOps Engineer Expert | Credential ID: 85A970745A50E145

Microsoft Certified: Azure Administrator Associate | Credential ID: 4C83D9DA3DD1BF20

RESEARCH & PUBLICATIONS

Liver Cancer Detection using Machine Learning IJRCSEIT | ijsrcseit.com/home/issue/view/article.php?id=CSEIT183818

ACHIEVEMENTS

- Gold medalist in chess at the Walmart Games in Bentonville, representing the Marketplace organization.